iAstroImageJ (AIJ) Tutorial for Transiting Exoplanet Data-Reduction & Analysis

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1. Introduction

AstroImageJ, or AIJ, is an ImageJ software that is customized for astrophysical purposes, mainly data reducing and analyzing images taken through optical telescopes. The main website for AIJ can be found at www.astro.louisville.edu/software/astroimagej/. This tutorial is intended to be a summarized version and geared toward data-reducing and analyzing transiting exoplanet data; more in-depth sources are listed near the end of this tutorial.

2. Installing, Updating, & Preparing AIJ

2.1. Installing AIJ

A list of installation packages can be found in www.astro.louisville.edu/software/astroimagej/installation_packages/. Note that Java is required to run AIJ; if one does not already have Java installed it will get installed concurrently when you install AIJ. To install AIJ, follow the directions below based on your computer's operating system:

1) For Windows users:
   a) Download the package via http://www.astro.louisville.edu/software/astroimagej/installation_packages/AstroImageJ_v5.3.3.00-windows-x86_64Bit.exe
   b) Extract the downloaded zip file and place the extracted folder in a directory that is convenient for you.

2) For Mac users:
   a) Download the package via http://www.astro.louisville.edu/software/astroimagej/installation_packages/AstroImageJ_v5.3.3.00-mac-arm_64Bit.dmg

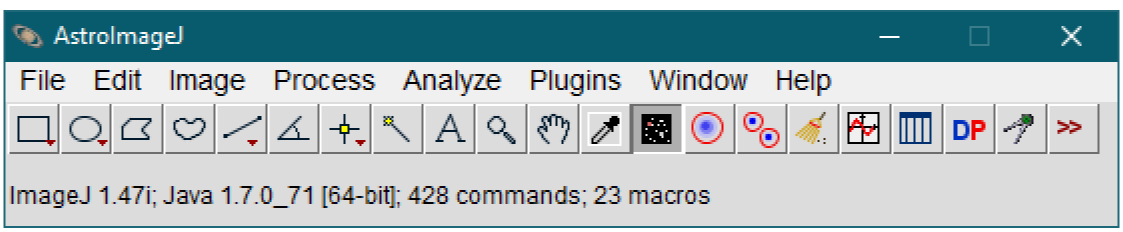
      And follow the instructions here www.astro.louisville.edu/software/astroimagej/installation_packages/AstroImageJ_installation_mac.html.
   b) Follow the prompted steps on installing the package.
   c) You may get a half-empty icon bar. To fix this, go to “Plugins” -> “Macros” -> “Install” (or press “command” + “shift” + “m”), navigate to the “AstroImageJ/Macros” folder, and run “StartupMacros.txt”. Additional tweaks are required to allow the macros to actually run. OSX 10.12 and 10.13 security “enhancements” are isolating scripts that should work together.

A more in-depth guide on AIJ installation can be found in Section 2 of “AIJ User Guide” pdf.

2.2. Updating AIJ

The next step is to update your AIJ package.

1) To run AstroImageJ, go to “AstroImageJ” folder and double-click on “AstroImageJ” icon. A small toolbar should pop up:



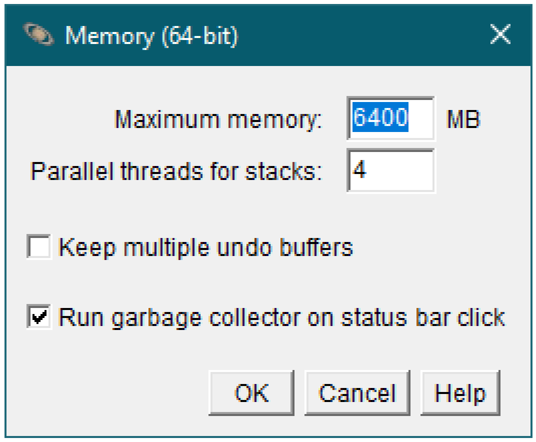
2) Go to “Help” -> “Update AstroImageJ…”. “AstroImageJ Updater” window will then open.

3) Click on the down arrow next to the displayed version number, scroll up, select “daily build”, and click “OK”. Your AIJ will be restarted and should then be updated to the latest beta version.

2.3 Preparing AIJ

Since many sets of images you will be data-reducing and analyzing will have very large number of bytes (often in gigabytes), you may need to adjust the memory allocation setting in AIJ.

1) On your AIJ toolbar, go to “Edit” -> “Options” -> “Memory & Threads…”. A window should appear:



2) The maximum memory will depend on the byte size of your RAM. As an example, for those with 8.00 GB RAM, change the “Maximum memory:” entry to roughly 6400 MB and click “OK”. If you enter too high of a maximum memory, your computer will experience memory thrashing which will considerably slow down your computer processes and may cause it to crash.

3) Close down and reopen AIJ for that change to take effect.

4) Go back to the memory setting and check if the new “Maximum memory:” entry is saved. If it was not, you probably entered too high of a value. Try making smaller changes, restart AIJ, then check again. You can repeat this step until you are satisfied with the max memory setting.

2.4 Priming AIJ for Plate-Solving

Plate-solving an image is defined as assigning each object in an image with RA and DEC coordinates. This is useful in that a set of plate-solved images do not need to be aligned before doing light curve analysis; this will greatly cut down the number of new files being created, which would save us more memory space in the long run. Another advantage is that if one wishes to align a stack of images, one can automatically start the alignment process without having to manually select the targets in each image. There are two options for doing plate-solving in AIJ, depending on your computer’s operating system:

1) For Windows users:
   a) Plate-solving can be done using a local version of Astrometry.net, which can considerably speed up the plate-solving process.
   b) Visit adgsoftware.com/ansvr/ to install an ansvr plugin.
   c) Follow the installation directions #1 through #14 on that webpage.
   d) Note that this plugin appears to be only available for Windows.

2) For Mac users:
   a) Plate-solving can be done using the online version of Astrometry.net.
   b) Register online at nova.astrometry.net/signin/?next=/api_help to obtain an Astrometry.net key. There is no fee for registration.
   c) Go to “My Profile” and look for “my API key”. You will need this key to initiate the plate-solving process on AIJ; entering the key will only need to be done once unless you reset the AIJ settings.

Section 7.4 of “Transit Light Curve Tutorial” pdf provides further discussions on preparing and using AIJ to plate-solve images.

3. AIJ Data-Reduction

A data-reduction is a process in which images are being rid of spurious and/or artificial counts caused by various external sources. There are four different types of images that are taken while observing a target of interest: bias, dark, flat, and science. In our case, only the darks and flats are needed to “clean” up the sciences before we can do any sort of analysis.

3.1 Bias

A bias is an image that is taken with a shutter closed and at an exposure time of nearly 0 s. This image captures systematic noises from the electronics within the camera. Normally, you would take at least 10 biases, median combine them into a master bias, and then subtract it from the raw sciences. However, doing so will add noise to the sciences. Fortunately, the bias subtraction does not need to be done since the darks already include the bias within them. Thus, the biases are not needed at all.

3.2 Dark

A dark is an image that is taken with a shutter closed and at an exposure time that is much longer than that of a bias. It contains noises generated through thermal processes. There are two sets of darks you should take: flat darks and science darks. Flat darks are darks with the same exposure time as the flats; these would be median combined into a master flat dark, which will then be subtracted from the raw flats. Science darks are darks with the same exposure time as the sciences; again, these would be median combined into a master science dark, which will then be subtracted from the raw sciences. You should take at least 10 darks for each set, so your overall total would be at least 20 darks. The darks are not dependent on the color filters used for the sciences.

3.3 Flat

A flat is an image that is taken with a shutter opened and at an exposure time such that it has enough counts (roughly 20,000) in each pixel. The flat is used to remove artificial distortions on the raw sciences caused by imperfect or defective pixels. A general rule-of-thumb is to take at least 10 flats, dark subtract each flat, median combine the flats into a master flat, and then flat divide the sciences by the master flat. The color filter for the flats should be the same one that is being used for the sciences.

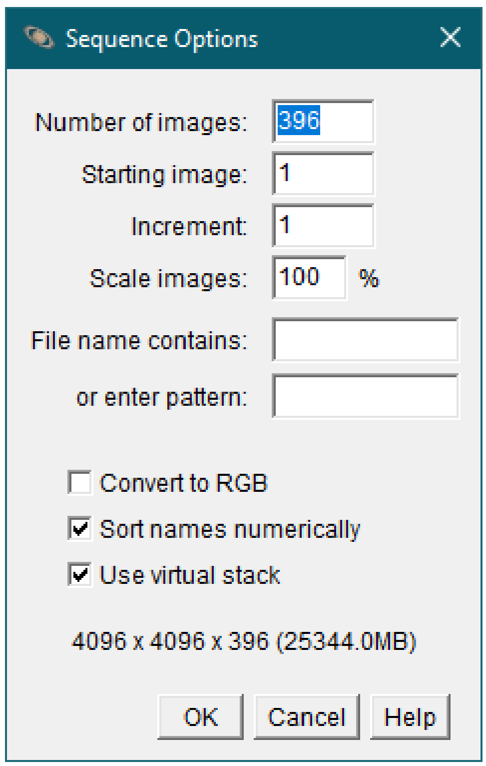
3.4 Science

A science is an image of a target you wish to analyze. Each set of sciences vary in number of images and exposure times. There are two types of sciences: A raw science is an unprocessed image while a reduced science is an image that has been data-reduced.

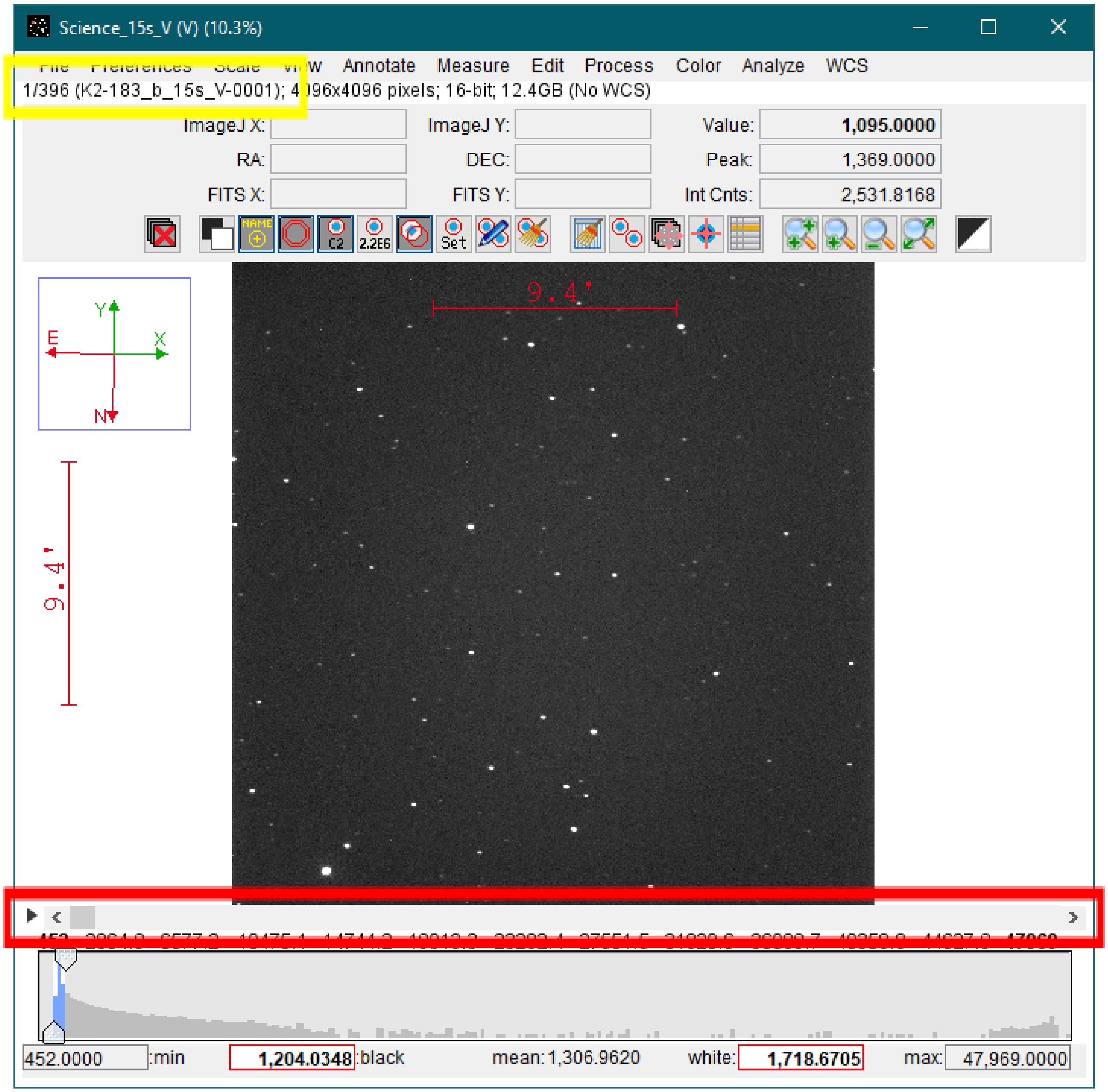
3.5 Preparing Files for Data-Reduction

1) During a given night, the observing volunteers would use the GMU Observatory telescope to take a set of images and save them into a directory folder labeled with the date of the observation. You would often be given a link to that folder.
   a) Download the images that are inside this folder (this process can take several minutes to even hours depending on the number of images taken).
   b) You must separate those images into the following groups (if it has not already done so): flat darks, science darks, flats, and sciences. The folder's label for each group should contain the type of image, exposure time, and the filter used. As an example: "Dark_8s", "Dark_15s", "Flat_8s_V", and "Science_15s_V".

2) Next, you must visually inspect each of your sciences for issues such as streaking, targets outside the field-of-view, or considerable shifting or jolting of the target between each frame (often due to the telescope periodically auto-correcting its position).
   a) Go to AIJ toolbar, click on "File" -> "Import" -> "Image Sequence…", then select the first science image file of a set that you wish to data-reduce; "Sequence Options" window will appear:



   b) Make sure the number of images is the same number of sciences that are part of a set and that "Use virtual stack" is checked. After clicking "OK", a stack of sciences will open up as seen on the next page.
   c) Near the top, just below the menu, are the sequence number out of the total number of stacked sciences and its corresponding file name (enclosed in a yellow box). Just below the image field is a horizontal scroll bar (enclosed in a red box) that allows you to page through each science; use that (or hit the right arrow button on your keyboard) to examine each science.

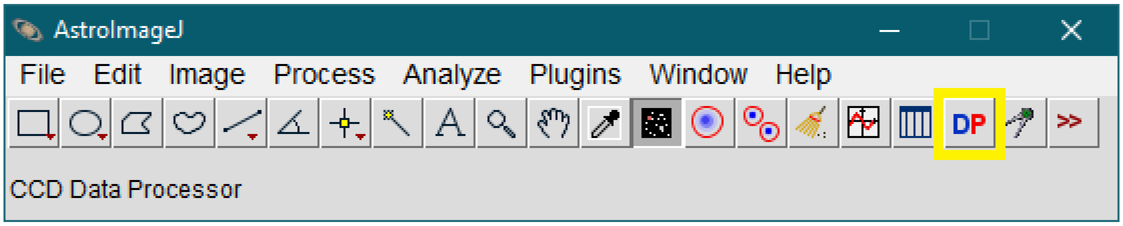


d) If you come across any bad images, create a new folder in the same directory with the dark, flat, and science folders, move those defective sciences into that folder, and create a text file within that folder with a list of science files you moved and the reason for the removal.

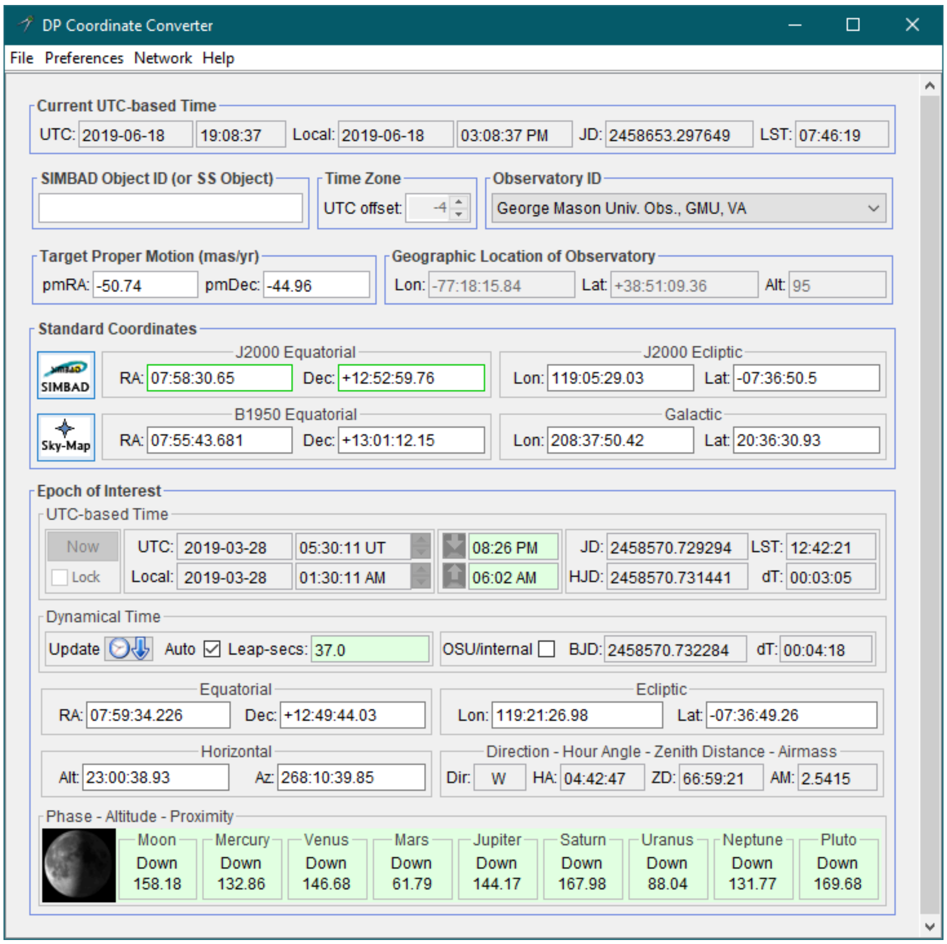
e) Once your set of sciences have been checked, you can then close the stacked images window and proceed to data-reduction.

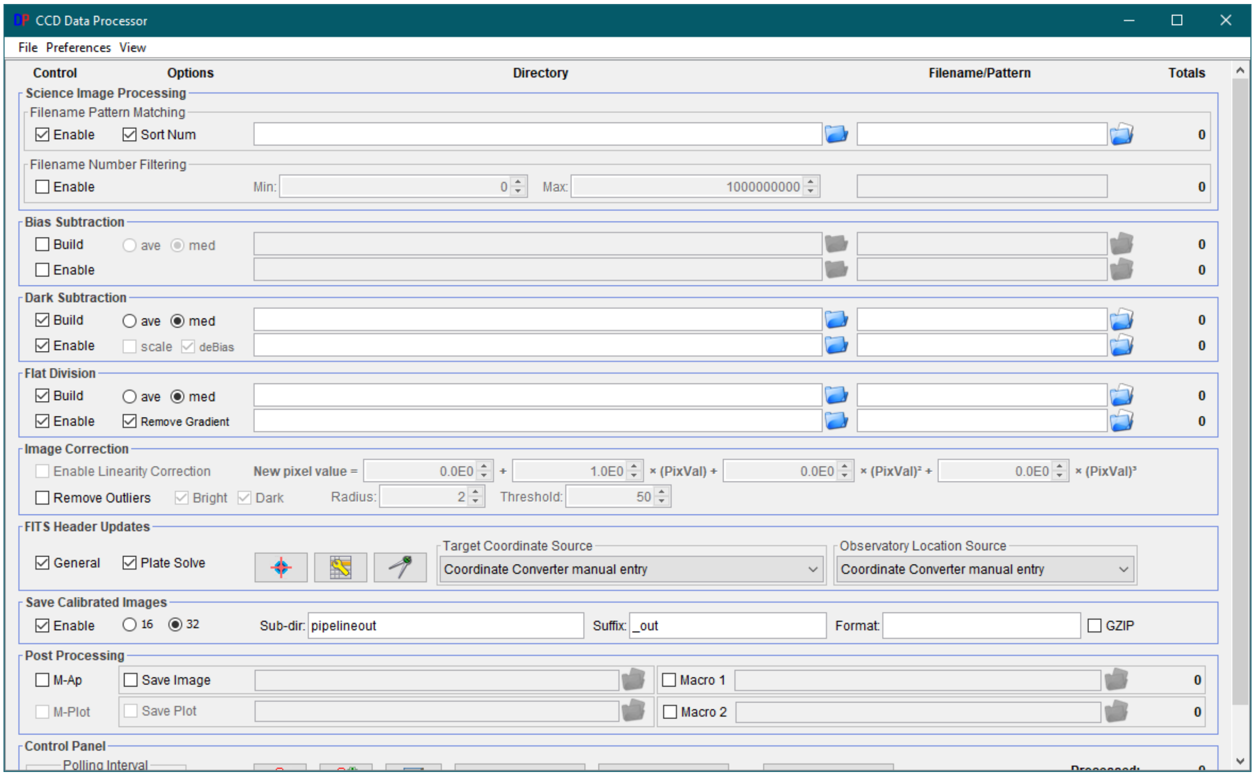
3.6 Doing Data-Reduction

1) To use AIJ to data-reduce your images, click on "DP" button (enclosed in a yellow box) on AIJ toolbar.



Two windows will materialize: “DP Coordinate Converter” and “CCD Data Processor”.





2) Go to “DP Coordinate Converter” window.

   a) Under the “Observatory ID”, make sure “George Mason Univ. Obs., GMU, VA” is selected; the “Time Zone” and “Geographic Location of Observatory” will be automatically filled. This only need to be done once, but if you reset the AIJ settings, you may have to repeat this step.

   b) You can enter the target’s name or ID into “SIMBAD Object ID (or SS Object)” to see if it does have info on that target. If it does, it will automatically enter the coordinates under “Standard Coordinates”. If it doesn’t, you will have to manually enter the target’s RA and Dec into “J2000 Equatorial” boxes; make sure you hit “enter” each time you fill in the box so that all other coordinate boxes are automatically updated.

   c) The “Target Proper Motion (mas/yr)” will often not be filled in. The target’s proper motions can be looked up in SIMBAD (simbad.u-strasbg.fr/simbad/), ExoFOP databases (exofop.ipac.caltech.edu), or NASA Exoplanet Archive (exoplanetarchive.ipac.caltech.edu) and typed into the boxes; again, hit “enter” each time you fill in the box.

   d) Under “Dynamical Time”, make sure “Auto” box is checked, then click “Update”. A window will open; click “OK” on that window. This step will update the leap seconds of your images’ timestamp in BJD (Barycentric Julian Date).

3) Go to your “CCD Data Processor” window.

   a) Note that “Bias Subtraction”, “Dark Subtraction”, and “Flat Division” have 4 boxes. The longer box on top specifies the location of your raw files. The longer box on the bottom specifies the output location of your reduced images. The shorter box on top allows you to enter a filename pattern, and the one on the bottom lets you specify the output master file name. You can hover your mouse above a box to see details about that input field.

b) The “Total” column on the far right will tell you how many files has been found; make sure it has found all of your appropriate files after you’ve filled in the boxes – no more, no less!

c) The first thing you should do is to dark subtract the flats and create a master flat.

   i) In “Science Image Processing”, uncheck the box next to “Enable” under “Filename Pattern Matching”.

   ii) In “Bias Subtraction”, uncheck “Build” and “Enable” (we don’t usually take biases).

   iii) In “Dark Subtraction”, check “Build” and “Enable” and select “med”. Click on the input row’s second blue folder icon; find and select one of the flat darks. (***IMPORTANT!*** You likely have two sets of darks – one has the same exposure time as your flats while the other has the same exposure time as your sciences. When you make your master flat and master dark here, you need to make sure that you are using your set of darks that has the same exposure time as your flats.) The directory address box and the filename box will be automatically filled in. Copy the directory entry and paste it into the output row’s directory box directly below it. In the output row’s filename box, type in “mdark_#s.fits” (replace # with the exposure time of those darks). This will create a master dark file with that name.

   iv) In “Flat Division”, check “Build” and “Remove Gradient”, uncheck “Enable”, and select “med”. Repeat step iii) for the boxes, except to type “mflat.fits” into the output’s filename box.

   v) Click on “START” button at the bottom. A log should come up that shows AIJ computing stuff. Let it run; it will say “Finished” when everything is completed.

d) The next thing is to data reduce the sciences.

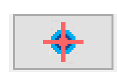
   i) In “Science Image Processing”, check “Enable” under “Filename Pattern Matching”. Click on the second blue folder icon to find the sciences you want to data reduce; select one of those files, and the boxes should be automatically filled in.

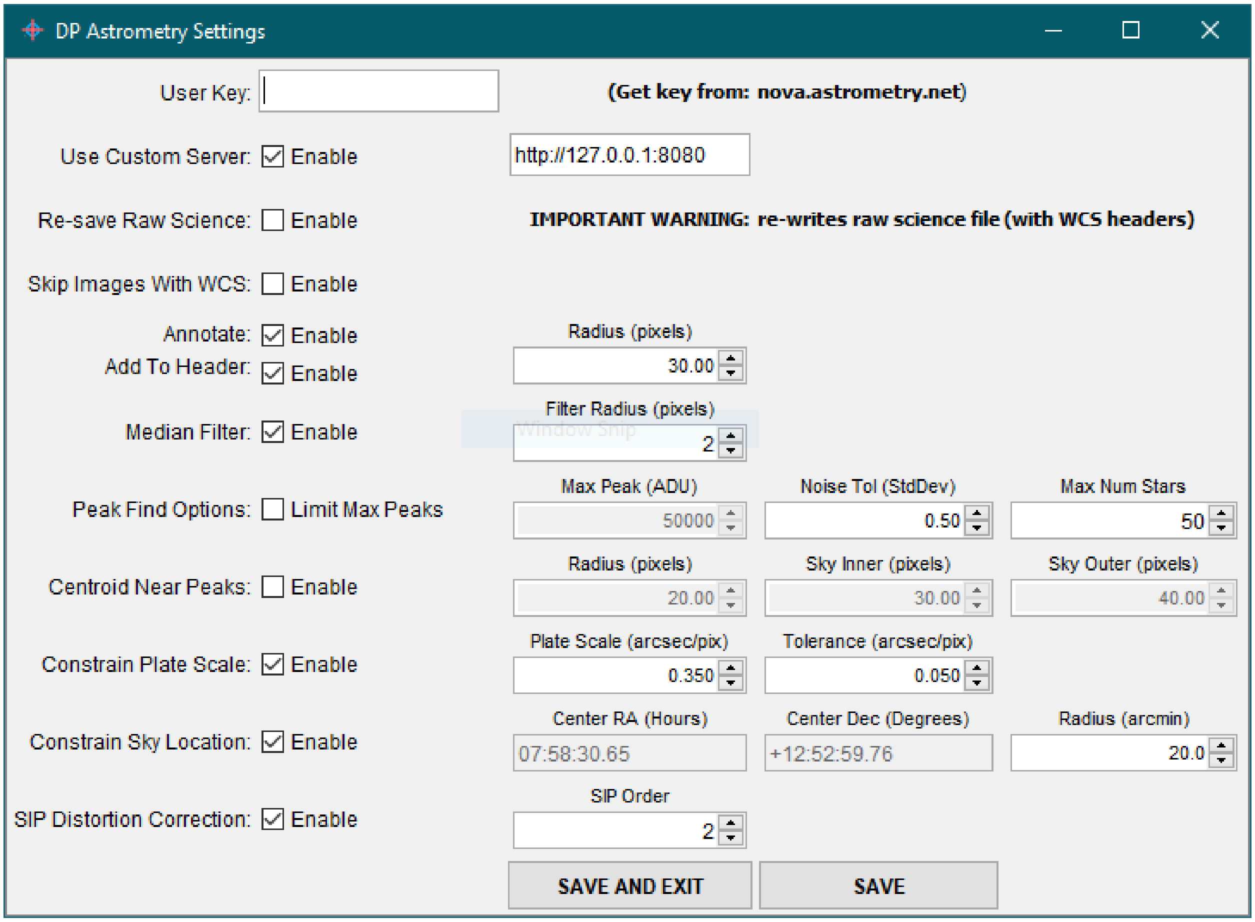
   ii) In “Dark Subtraction”, repeat step c iii) except to select one of the science darks. Make sure you change the exposure time number in the output filename box and to update the output directory box.

   iii) In “Flat Division”, uncheck “Build” and check “Enable”.

   iv) In “FITS Header Updates”, check “General” and “Plate Solve”. Make sure “Coordinate Converter manual entry” is selected for both “Target Coordinate Source” and “Observatory Location Source”.

   v) In “Save Calibrated Images”, check “Enable” and select “32” bits. The “Sub-dir:” box is to name a subdirectory folder that will be created to store the reduced sciences; the default name is “pipelineout”. The “Suffix:” box is to add a suffix to the reduced sciences’ filename to distinguish them from the raw science files; the default suffix is “_out”.

   vi) Go back to “FITS Header Updates”. Click on the plate solve icon . “DP Astrometry Settings” window will appear:



vii) For Windows users, leave "User Key:" box blank and check "Use Custom Server:". The box next to that should be automatically filled in like the one above.

viii) For Mac users, enter your API key into "User Key:" box and uncheck "Use Custom Server:". This will only need to be done once, but you may need to repeat this step if you reset the AIJ settings.

ix) Have all other boxes be exactly those shown in the snapshot above. Also, make sure the entries for "Radius (pixels)", "Filter Radius (pixels)", "Plate Scale (arcsec/pix)", "Tolerance (arcsec/pix)", "Radius (arcmin)", and "SIP Order" are the same. Also, the "Center RA" and "Center Dec" boxes should already have entries; those were updated via "DP Coordinate Converter" window. Click "SAVE AND EXIT".

x) In "FITS Header Updates", click on the icon . "General FITS Header Settings" window will emerge. Find the "BJD (TDB) mid-Obs Keyword:" box. The default entry is "BJD_UTC"; change that entry to "BJD_TDB", then close out that window. This only need to be done once, but this may need to be repeated if you reset the AIJ settings.

xi) Click "START" and let it run until the log says "Finished". Your reduced and plate-solved sciences should now have been created in the new subdirectory "pipelineout".

xii) You can choose not to plate-solve the sciences if you want to speed up the data-reduction process. Simply uncheck "Plate Solve" in the "CCD Data Processor" window before running the data-reduction.

4. Alnitak Data Reduction & Plate-Solving

5. AIJ Light Curve Extraction

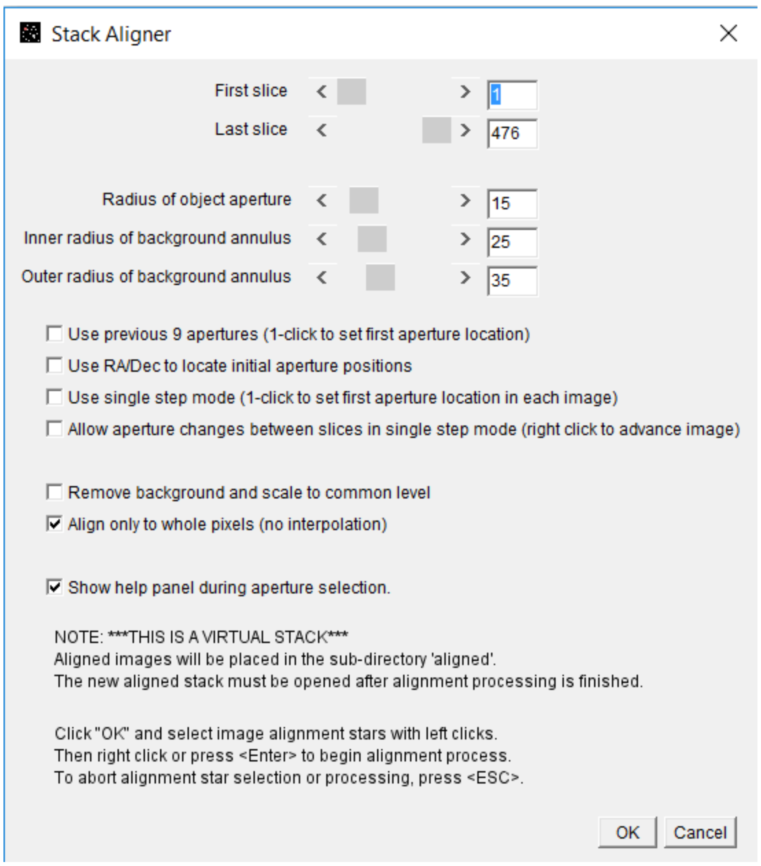
5.1 Aligning Images

If you did not plate-solve the sciences, you will need to align them before doing light curve extraction.

1) Import a virtual stack of data-reduced sciences. See Section 3.5.2a and b in case if you have forgotten on how to do that.

2) Page through the stack and note whether or not there are large “jumps” (i.e., is the star in one place in one image, and then in the next it’s in a very different, far away place?). This will be important in a moment.

3) In the stack window (not AIJ toolbar), go to “Process” -> “Align stack using WCS or apertures…” A window will come up that looks like this:



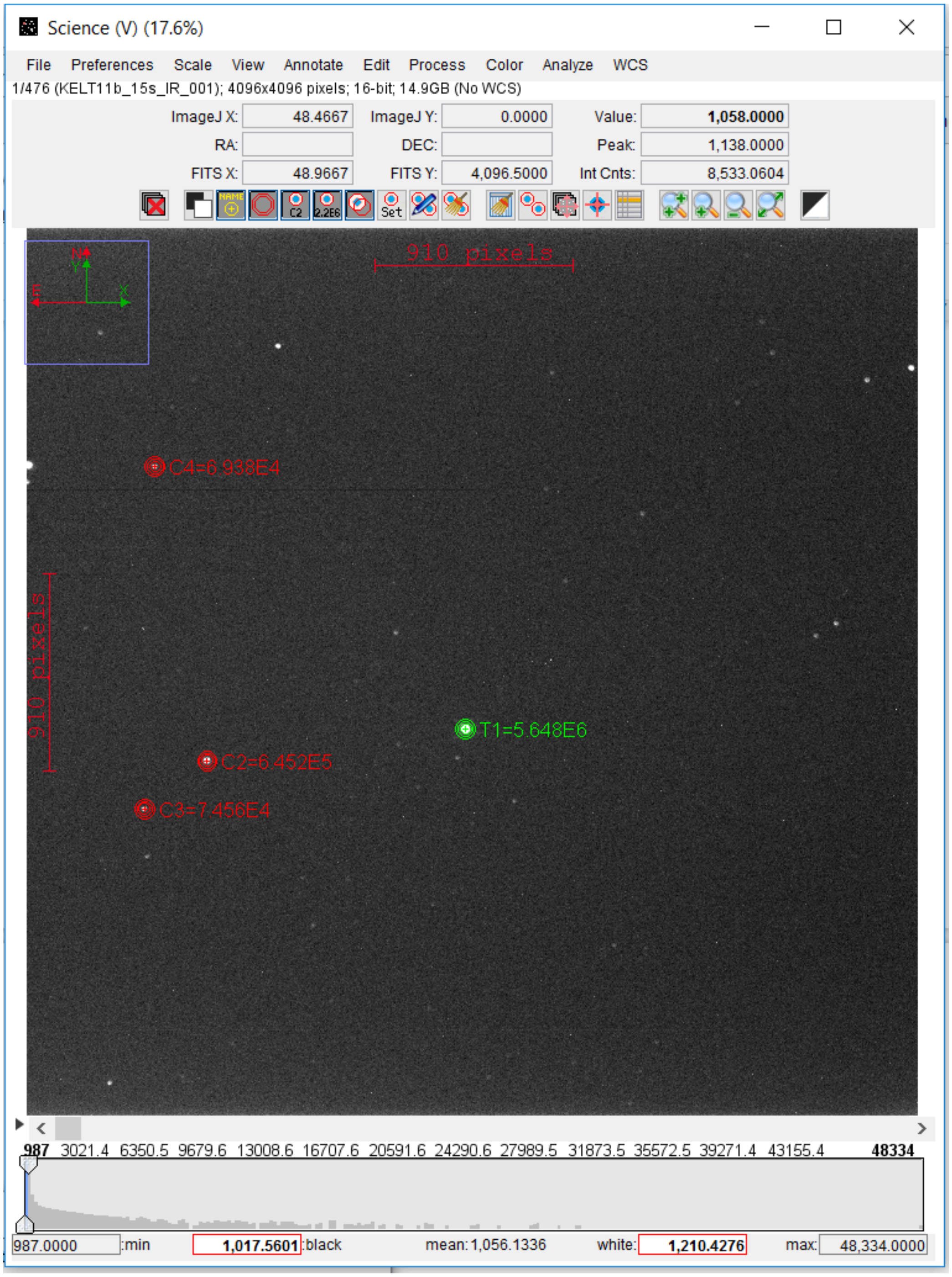
a) Make sure you select all the slices in your stack. Set “Radius of object aperture” to about the size of your target star. You can just add 10 (as in the above image) to get the other aperture sizes. There is a rigorous way to choose apertures; see Section 4.2.3 if you wish to employ that method.

b) Have all checked boxes be like those in the screenshot above. Note: "Use previous n apertures" will just use the same aperture size, location, and number you used last time, so keep it unchecked for this purpose.

c) If you noticed significant jumps in your stack, check "Use single step mode".

d) Click OK.

4) Now, you can place your apertures. The first one you place will always indicate your target star (marked as T1 in green). The others are reference stars (marked as C# in red). For alignment of images, you only need 3 or more reference stars. Things should look something along the lines of this:



5) Once you have all your reference stars selected, you can right click on the image or hit "enter".
   a) If you did not enable "single step mode", life is good and it will align all the images for you.
   b) If you had to enable "single step mode", you will have to click on your target star for each science you have. This is arguably the worst part, but at least you don't have to click your reference stars over again!

c) Aligned images, now denoted with a filename suffix "_aligned", should be placed in a newly-created folder called "Aligned", which should be located in the same folder your processed images are in.

5.2 Generating a Measurement Table for Light Curve

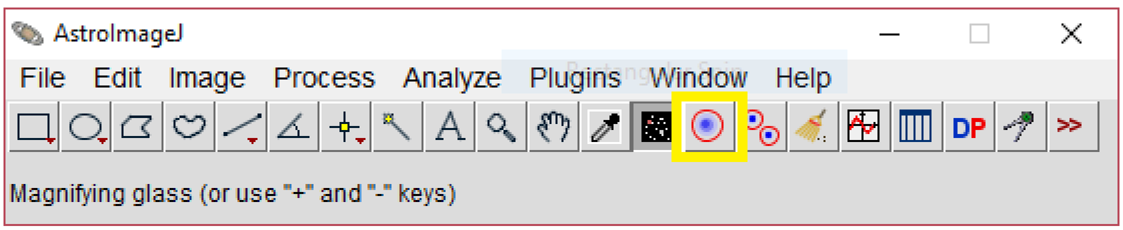
We can now generate a light curve using our plate-solved or aligned data.

1) Please review Section A (pages 3 and 4) of TFOP SG1 Guideline pdf for file naming conventions before proceeding to light curve extraction. From this point on, you will be saving files that are needed as part of submitting the report online.

2) Import a virtual stack of data-reduced and plate-solved (or aligned) sciences. See Section 3.5.2a and b in case if you have forgotten on how to do that.

3) We need to figure out the aperture size so that we can do the aperture photometry properly.

   a) On AIJ toolbar, click on "Aperture Photometry Tool" button (enclosed in a yellow box).

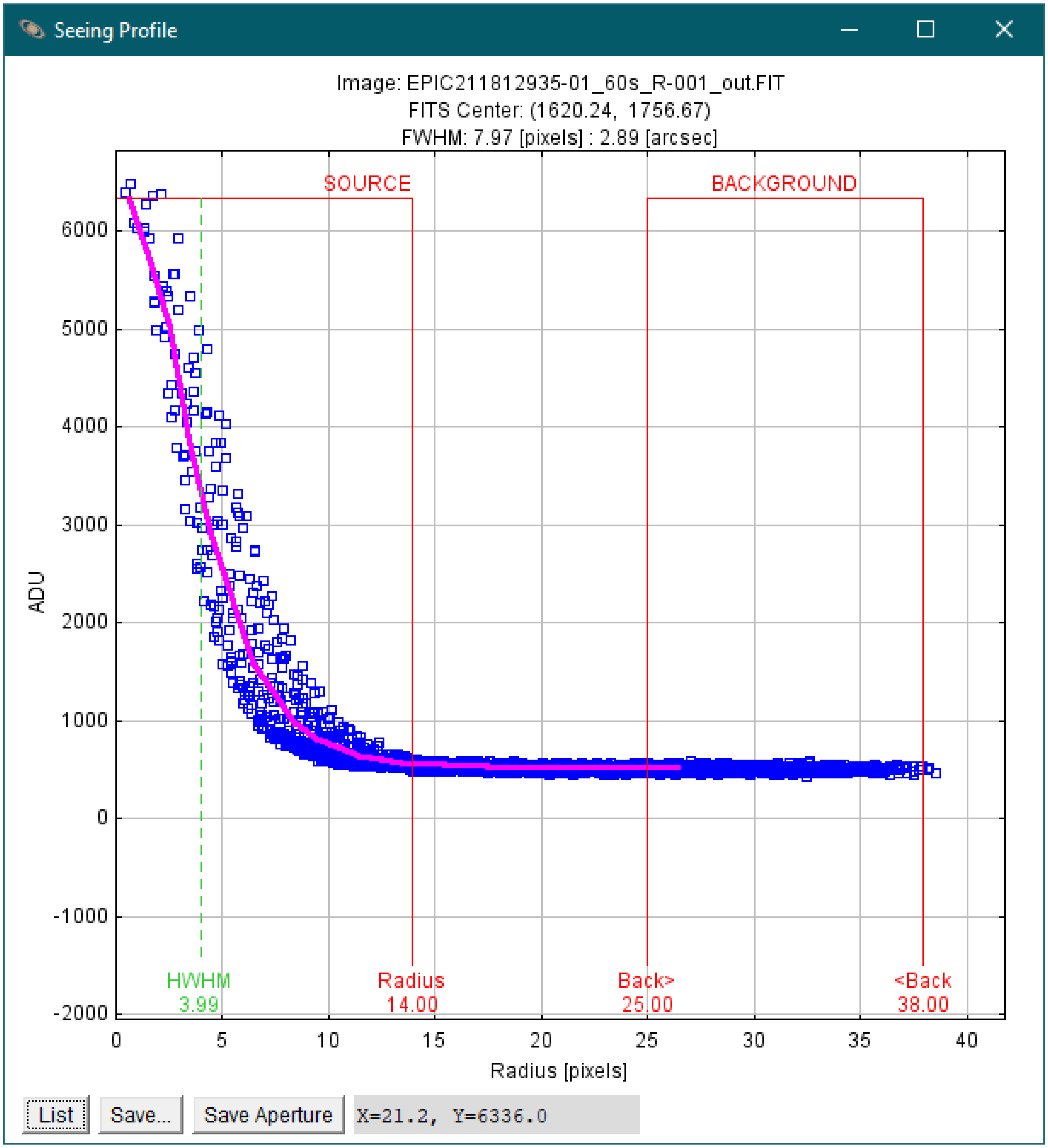


   b) Use the provided finder chart to find your target on the first science image. Once you have found your target, click on it. "Measurement" window may pop up; close it. There should be one or three red circles enclosing the target now.

   c) On AIJ toolbar, go to "Plugins" -> "Astronomy" -> "Seeing Profile". "Seeing Profile" window will materialize.

      i) The source's radius will be your aperture size, and the background's inner and outer radii will be your annulus sizes.

      ii) Save the seeing profile plot by clicking on "Save…" button near the bottom left corner. Once you have saved the file, close that window.



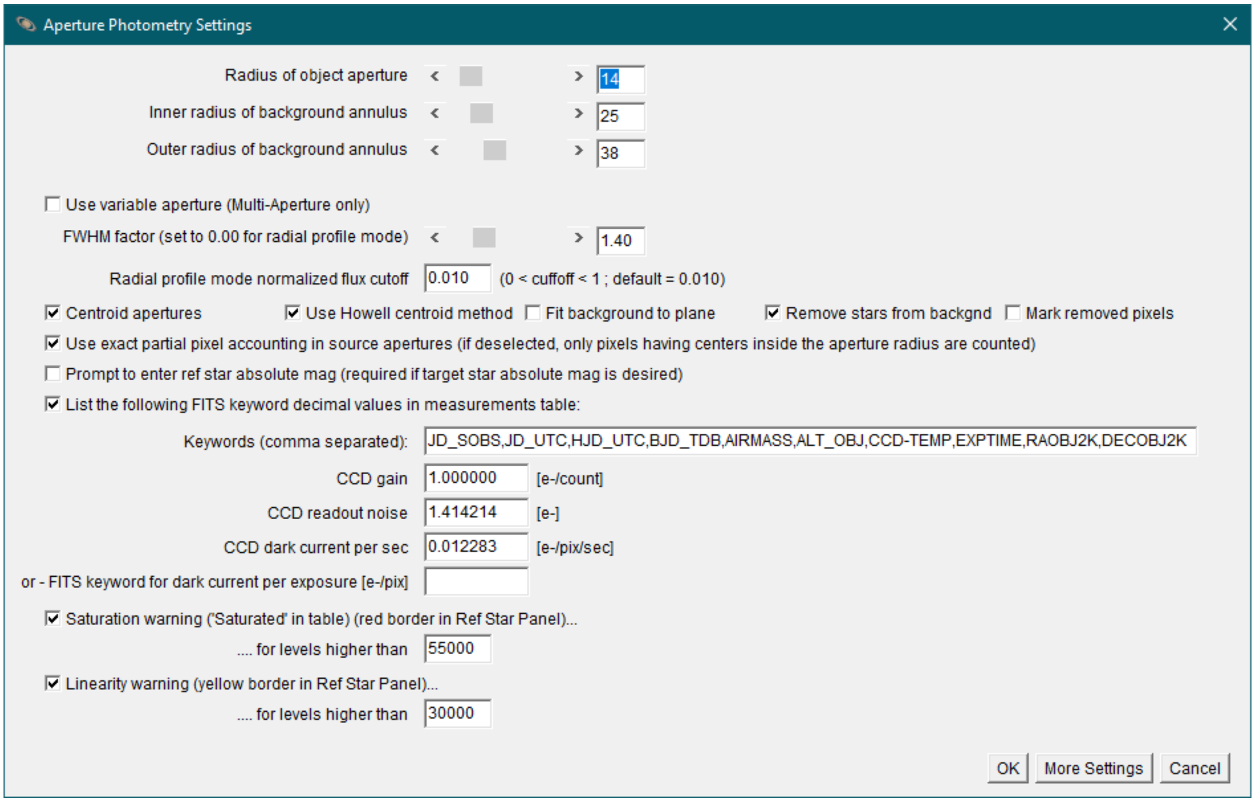
d) On AIJ toolbar, double-click on "Aperture Photometry Tool" to open "Aperture Photometry Settings" window.

i) Enter the aperture and annuli radius values you obtained from the seeing profile plot.

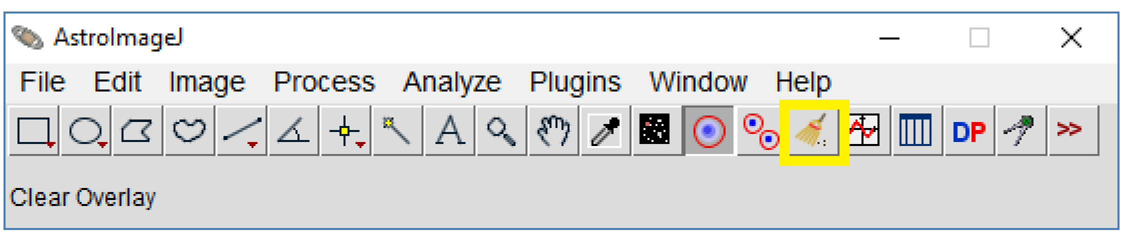
ii) All other entries and boxes should be exactly like those in the screenshot. You may need to enter "1.414214" into "CCD readout noise" entry box and "0.012283" into "CCD dark current per sec" entry box; those are based on the specifications of our camera's CCD and should only need to be entered once unless you reset the AIJ settings.

iii) After you are done with the settings, click “OK”.

iv) Step 3d can be skipped if “CCD readout noise” and “CCD dark current per sec” entries are already updated. The aperture and annuli radius values can instead be entered via Multi-Aperture Photometry (see step 4).

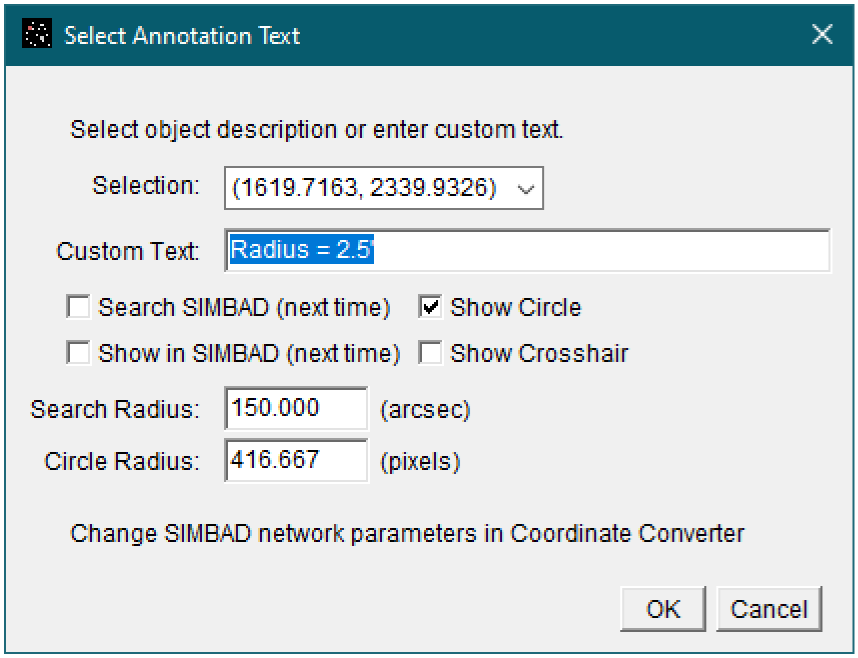


e) On AIJ toolbar, click on the “Clear Overlay” button (enclosed in a yellow box). This will remove the red circles on the image.



4) Now, we should be ready to do multi-aperture photometry.

a) We need to temporarily place a 2.5’ circle around the target. In the image stack window, right-click on the target. The following window will emerge:



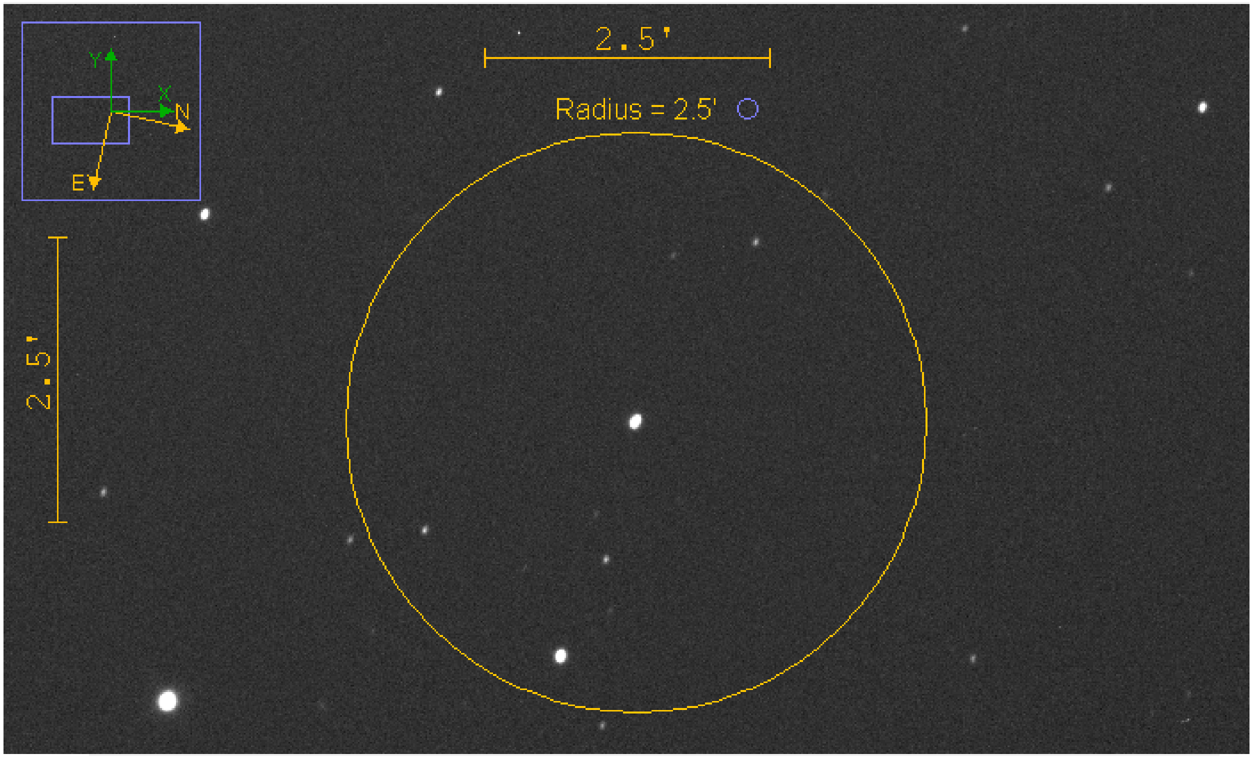
i) In "Custom Text:" entry, type in "Radius = 2.5′".

ii) Check "Show Circle" and uncheck all other boxes.

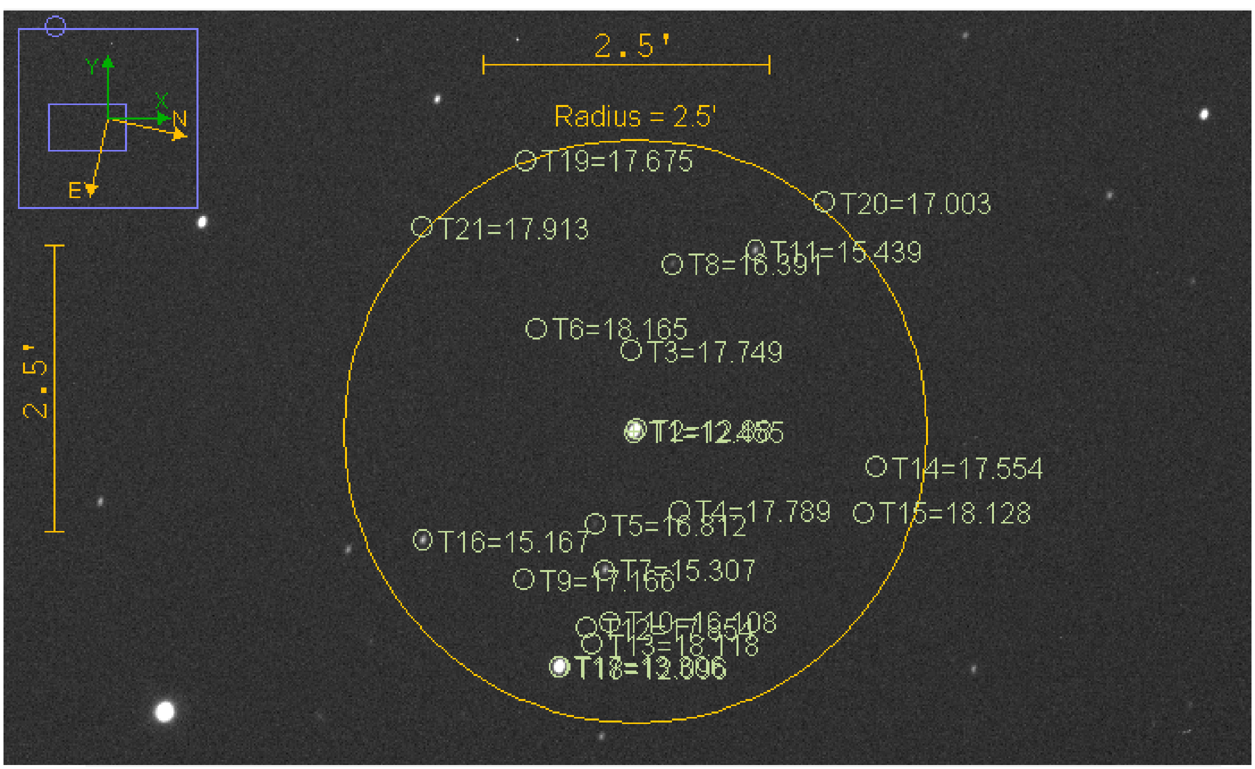
iii) In "Search Radius:", type in "150.000"

iv) In "Circle Radius:", type in "416.667". This is based on the specifications of our CCD.

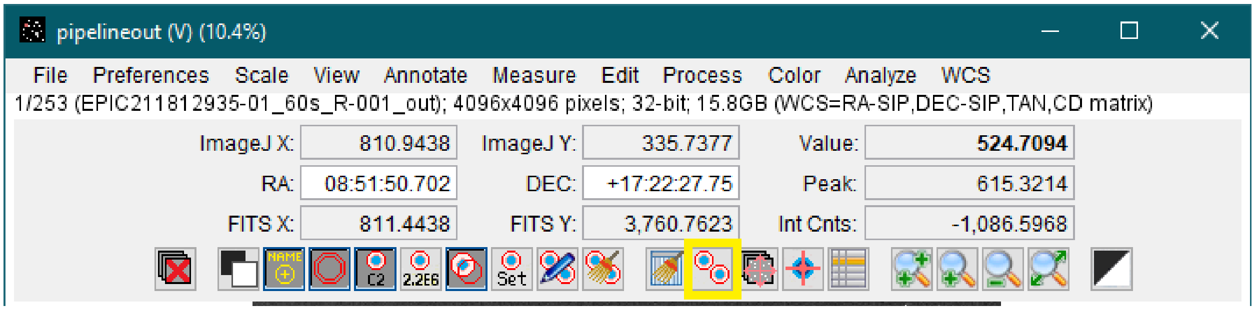
v) Click "OK". An orange circle will appear in your image:

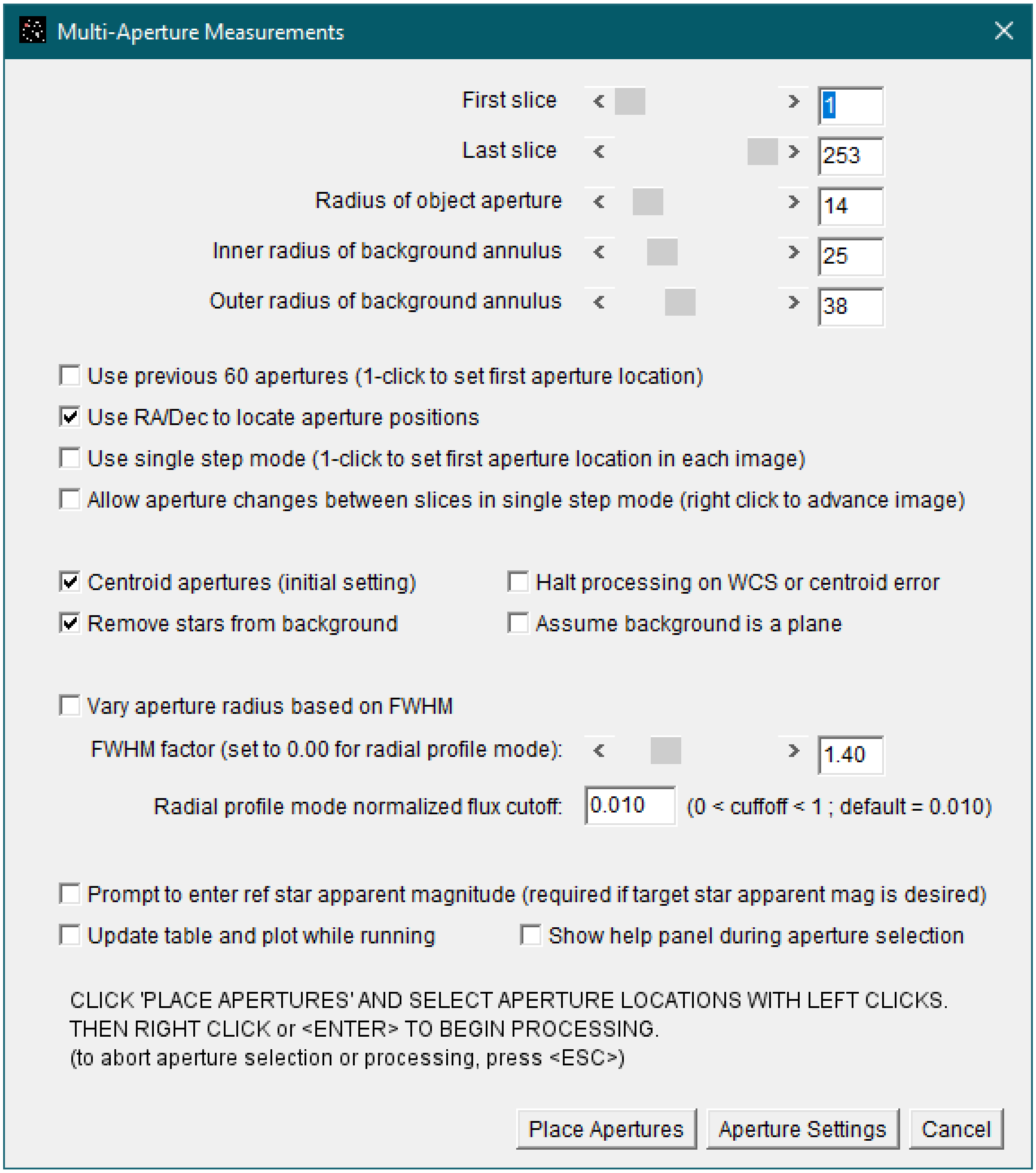


b) If you have a Gaia stars .radec file that was generated for that particular date in which this target's data were collected, you can drag that file over the image and drop it. You should then see something like this:



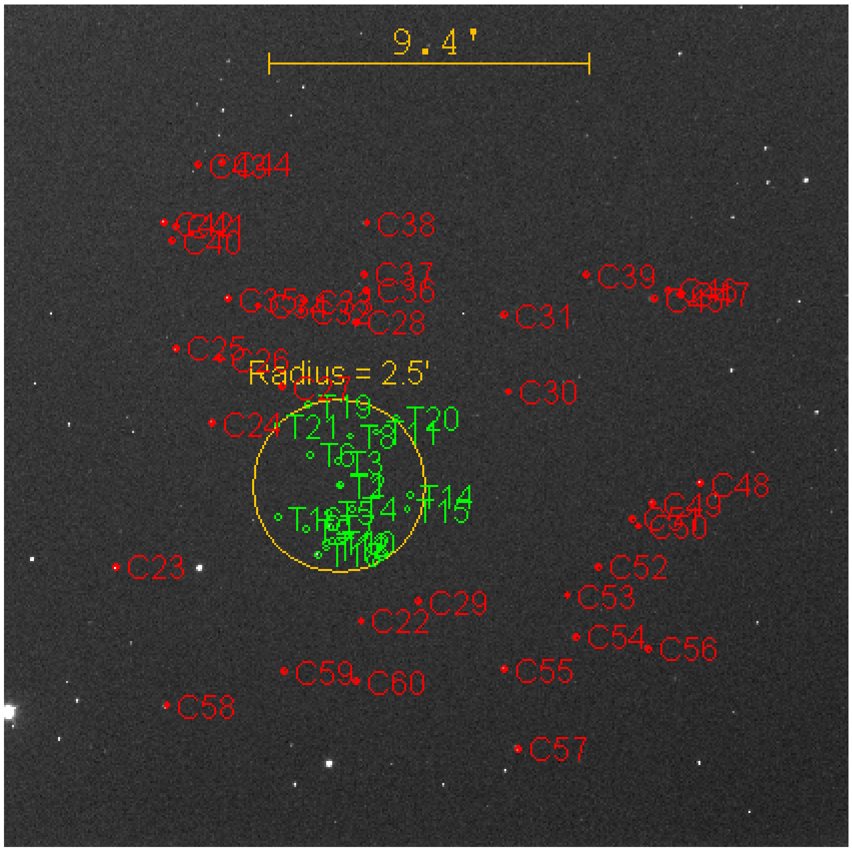
c) In the image stack window, there is an icon with two red circles with a blue dot in the middle of each circle (enclosed in a yellow box). If you hover over it, it will say “perform multi-aperture photometry”. Click it to bring up “Multi-Aperture Measurements” window:





i) If you skipped step 3d, enter the radii for aperture and annuli using the values from the seeing profile plot.
ii) If your set of images are plate-solved, check “Use RA/Dec to locate aperture positions” and uncheck “Halt processing on WCS or centroid error”.
iii) If your set of images are not plate-solved, uncheck “Use RA/Dec to locate aperture positions” and check “Halt processing on WCS or centroid error”.
iv) Have all other boxes be kept as seen in the screenshot above.
v) Click “Place Apertures” at the bottom of the window.

d) You will first select your target star, followed by your reference stars. Unlike with alignment, you need as many good quality reference stars as you can get. Try to get around 10 or so. Good reference stars are of the same size and brightness of your target star. If you can't get many good matches, that's OK. Just take what you can get. However, you may want to avoid choosing reference stars near the edges as they may have drifted out of sight in later images. Note: If you did step b, the target stars may already have been selected; you may only need to select reference stars around them. As an example, your resulting selections may look like this:



e) Once your apertures are placed, you may want to save the selections as a backup. Go to "File" -> "Save apertures..."; an aperture file will be created. That way, this can cut down your effort and time if you need to redo the multi-aperture photometry.

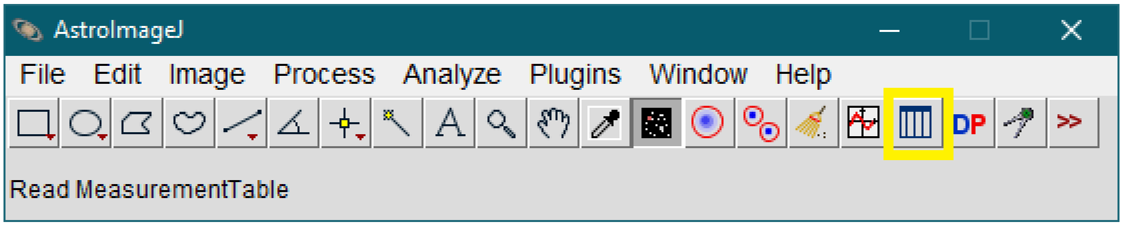
f) Once you are done with the above steps, right click or hit "enter". The "Measurements" window will pop up. It will remain blank until the process is completed; let AIJ finish processing before you play around with anything.

g) Once the process is completed and the "Measurements" window is now populated with data, go to that window and click on "File" -> "Save As…". You now have a measurement table that you will need to create a light curve plot.

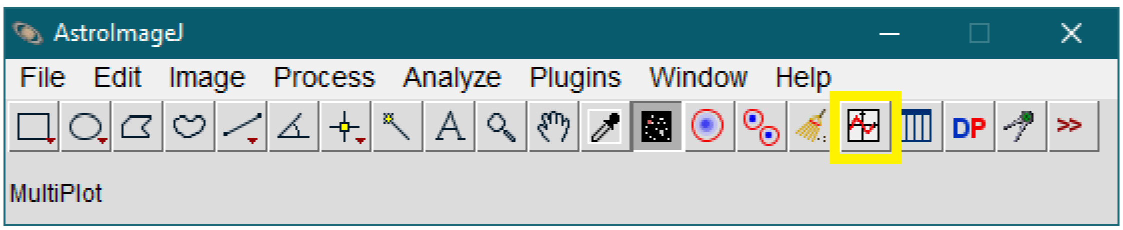
6. Analyzing & Interpreting Light Curves

In this section, you will be analyzing and then interpreting the light curve.

1) If you don't have the measurement table up right now, click on the measurement table button (enclosed in a yellow box) on AIJ toolbar, then find the "..._measurements.xls" file that you saved in the previous steps. This will open up the "Measurements" window.



2) To do the light curve analysis, click on the multiplot button (enclosed in a yellow box) on AIJ toolbar. Multiple windows will then materialize.



3) The first thing you should do is to upload the default template.

   a) If you don't have "Template.plotcfg", download it via www.astrodennis.com/Template.zip, unzip the file, and place it in a convenient directory.

   b) Go to "Multi-plot Main" window. Click on "File" -> "Open plot configuration from file…". Find and select "Template.plotcfg". This will revert all plotting windows to proper settings so that your analysis will be in compliance with the TESS/K2 reporting guidelines. You should do this every time you start a new light curve analysis.

4) In "Multi-plot Main" window, do the following steps:

   a) Set the "Default X-data" to "BJD_TDB".

   b) In "V. Marker 1" and "V. Marker 2" boxes, enter the predicted ingress and egress times. You can get these values by consulting the target's "Transit Info" png file and looking up the "BJD_{TDB}" column. Only take the decimal portion of the time values; the start and end times should be in 0.### format (number of decimal places may vary from target to target). If, for example, the start time was "8191.630" and the end time was "8191.744", the entries should be "0.630" and "0.744". If the end time was "8192.744" (effectively the next BJD_TDB day), then the entry should be "1.744".

   c) In the "Title" section, enter the target's name and the UT date of observation.

   d) In the "Subtitle section, replace "Observer Name" with "GMU" and enter the filter used and the length of exposure time.

e) In the “X-Axis Scaling” section, select “Auto X-range” so that your data will become visible in the plot window.

f) In the “Fit and Normalize Region Selection” section, click on “Copy” icon. This will copy the entries from “V. Marker 1” and “V. Marker 2”. You will come back to this section for adjustments later on.

g) If your plot (in a different window) is too big that part of it is being obscured, go to the “Plot Size” section and reduce the “Height” and/or “Width” values until you can see the entirety of the plot window.

5) Go to “Data Set 2 Fit Settings” window and do the following steps:

   a) In “Orbital Parameters”, enter the target’s period. The value can be obtained via the target’s “Transit Info” png file.

   b) In “Host Star Parameters”, enter the stellar radius of the target’s host star. The value can be obtained from ExoFOP databases (exofop.ipac.caltech.edu) or NASA Exoplanet Archive (exoplanetarchive.ipac.caltech.edu). The rest of the parameters will be automatically updated.

   c) In “Transit Parameters”, you will need the “Linear LD u1” and “Quad LD u2” values.

      i) Visit astroutils.astronomy.ohio-state.edu/exofast/limbdark.shtml, choose the filter band that was used for your target, and enter the host star’s effective temperature, metallicity, and surface gravity parameters. Those values can be found in ExoFOP databases (exofop.ipac.caltech.edu) or NASA Exoplanet Archive (exoplanetarchive.ipac.caltech.edu).

      ii) If the metallicity is not provided, you can enter “0” to treat it as having metallicity like our Sun.

      iii) If either effective temperature or surface gravity parameter is not provided, have u1 and u2 be set to 0.3, unlock u1, and lock u2.

   d) In “Detrend Parameters”, have all parameters be unchecked. We don’t want to apply any detrending yet at this point.

   e) In “Plot Settings”, check “Show Residuals” and “Show Error”. The residuals and their error bars will appear on the plot.

6) Go to “Multi-plot Y-data” window and do the following steps:

   a) Review the label of each column. You can use the scroll bars at the bottom and at the right to move around.

   b) Make sure the “Plot” boxes are checked for the following “Y-data”:

      i) Sky/Pixel_T1

      ii) Width_T1

iii) AIRMASS

iv) tot_C_cnts

v) X(FITS)_T1

vi) Y(FITS)_T1

c) Page 8 of "TFOP SG1 Guidelines" pdf lists the recommended scale, shift, and color of those parameters listed in the previous step; have these settings of each parameter be exactly those listed and make sure each parameter's "Page Rel" box is checked.

d) We now need to review each reference star's flux.

i) Find "rel_flux_T2" or "rel_flux_C2"; it is usually in row 3. Check "Plot" and then examine this star's flux on the plot.

ii) If the flux shows large variation or scattering, make a note of that star (on a sheet of paper or on the computer's notepad); you will be removing that star from the list.

iii) Repeat steps i and ii for all other reference stars. You can change the "Y-data" entry to review each star instead of going down the rows.

iv) After reviewing each star's flux, you can uncheck the "Plot" box.

7) Go to "Multi-plot Reference Star Settings" window and do the following steps:

a) Uncheck all reference stars that you listed as bad from the previous step.

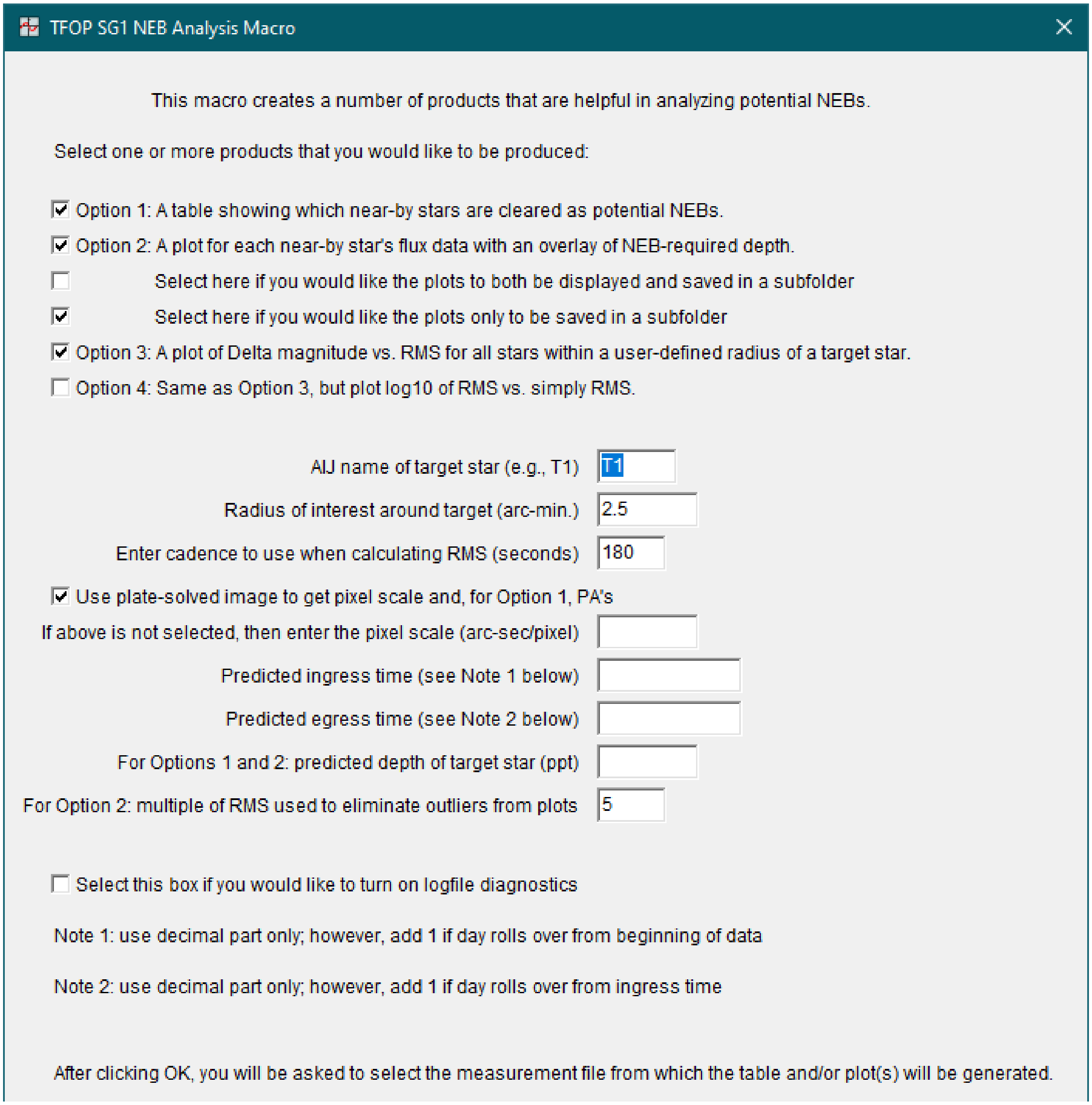
b) Note: "T" means the star is not being used as a reference star while "C" means the star is being used as a reference star.

c) Click on "Save Apertures" button to save the current aperture selections. If you had previously created an aperture file while doing multi-aperture photometry, you can overwrite that file.

d) Click on "Save Table" button to rewrite the measurement table file with the current aperture configurations. That way next time when you reload the measurement table and plot for later review, the aperture configurations will be exactly as you left it.

8) If you have already uploaded the .radec file prior to running the multi-aperture photometry, we'll need to check the neighboring stars to see if they're cleared or not.

a) Go back to "Multi-plot Main". Go to "File" -> "Create NEB search reports and plots…" The "TFOP SG1 NEB Analysis Macro" window will appear:



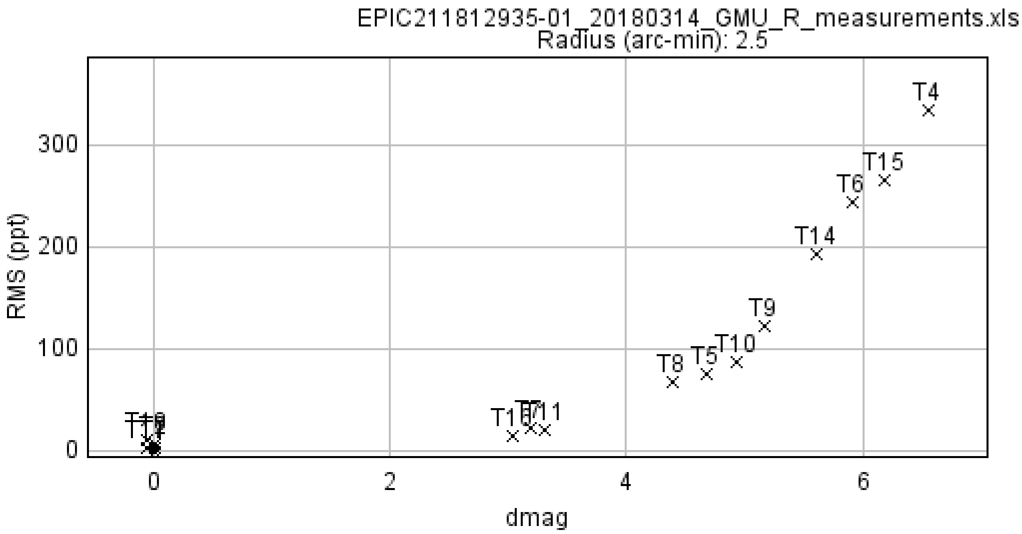
b) Have “Option 1”, “Option 2”, “Option 3”, and “Select here if you would like the plots only to be saved in a subfolder” be checked.

c) If your images are plate-solved, check “Use plate-solved image to get pixel scale…”

d) If your images are not plate-solved, enter “0.36” into the box below instead.

e) The predicted ingress and egress times and predicted depth can be obtained from the target’s “Transit Info” png file; enter those values into the boxes.

f) Click “OK” at the bottom. You will be prompted to select the measurement table file (and a plate-solved image, if you checked the box in step c). The “...dmagRMS-plot.png”, “...NEB-table.txt”, and a folder with multiple plots will then be generated.



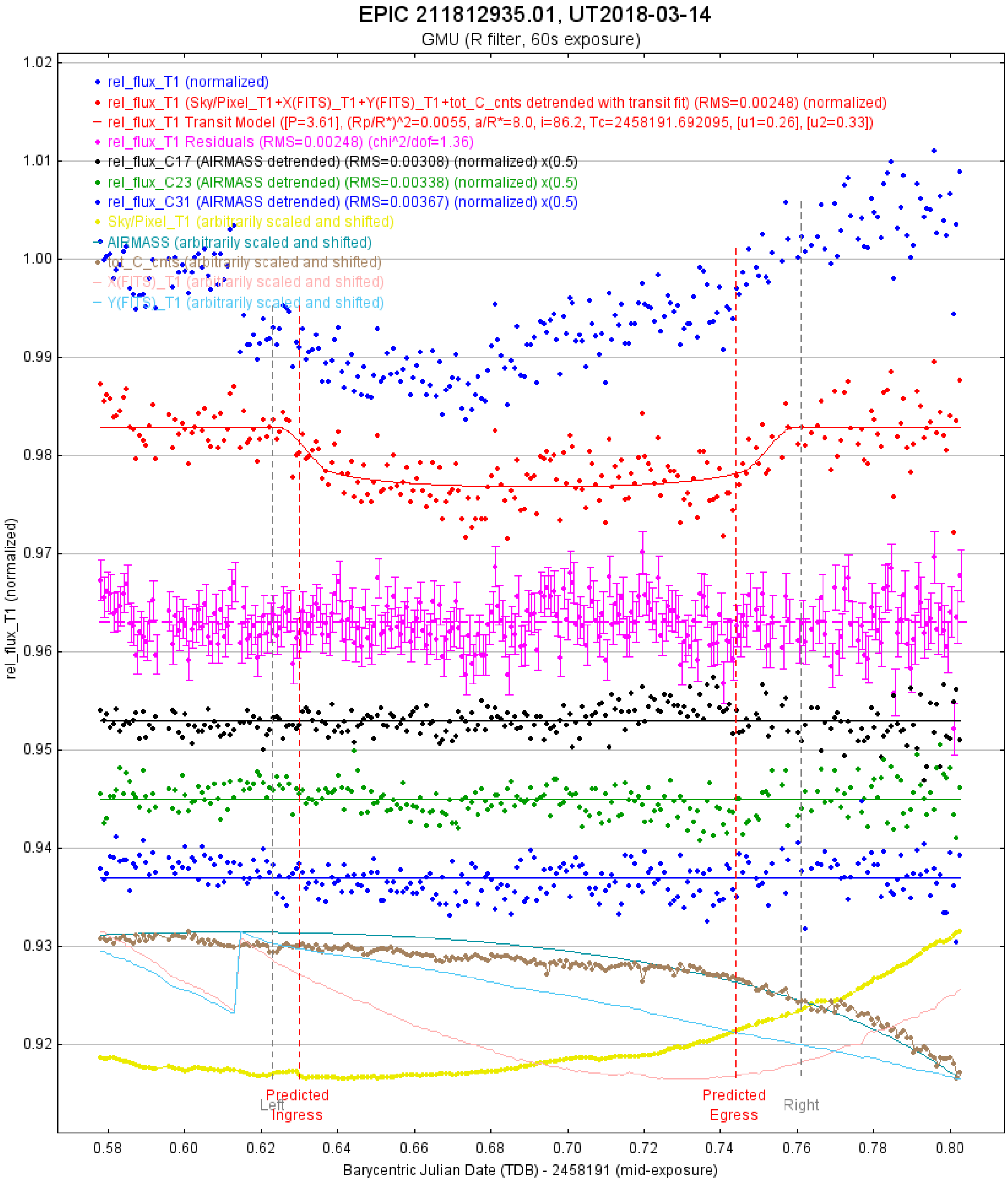
g) Examine the dmagRMS-plot for any outliers. If there are outliers, go to “Multi-plot Reference Star Settings” and uncheck those outliers (if they’re not already unchecked).

h) In “Multi-plot Reference Star Settings” window, click on “Save Table” and “Save Apertures” to resave the table and aperture files.

9) Examine the plot. If you see the T1’s transit, check the alignment of the predicted ingress and egress lines. If they’re not aligned, go to “Multi-plot Main” window and to “Fit and Normalize Region Selection” section; adjust “Left” and “Right” values until the grey lines match up with the ingress and egress on the plot.

10) Go to “Data Set 2 Fit Settings” window, check a detrend parameter “Use” box, then examine the plot. Do this for each detrend parameter one at a time.

11) At this point, interpreting the data is an open-ended question. Experiment with the detrending parameters and plot settings (“Multi-plot Main” and “Multi-plot Y-data”) until you believe you have a satisfactory result. Your resulting plot may look somewhere along the line like this:



12) The analysis steps may vary depending on what objectives the team is requesting (verifying the target's transit, checking the field for NEBs, etc). See "TFOP SG1 Guidelines" pdf for additional details.

7. Preparing Files for ExoFOP–TESS or ExoFOP–K2

Once your analysis is complete, you will need to prepare several files and write a brief summary about your results. Pages 3 and 4 of "TFOP SG1 Guidelines" pdf lists the required files for submission and their naming convention. Also consult that pdf file for additional details as the report preparations may vary depending on the objectives.

1) Create a new subdirectory folder that will contain your submission report files. Make sure to move and/or save all submission report files into that folder.

2) You should already have a seeing profile file saved (see Section 4.2.3).

3) You should already have a measurement table file saved with the latest aperture configurations. If you have not done so yet, go to "Multi-plot Reference Star Settings" window and click on "Save Table" button.

4) Save your current plot configurations by going to "Multi-plot Main" window and clicking on "File" -> "Save plot configuration…"; this will generate a .plotcfg file.

5) Go to the window containing the plot and click on "Save…" button at the bottom; this will save your plot as a light curve png file.

6) You should already have an aperture file saved with the latest aperture configurations. If you have not done so yet, go to "Multi-plot Reference Star Settings" window and click on "Save Apertures" button.

7) Copy your first plate-solved image fit file, paste it into your submission report folder, and rename that file according to TFOP SG1's naming convention.

8) If your plate-solved image is not up yet, double-click on that fit file to open it. Go to "Multi-plot Reference Star Settings" and click on "Show Apertures" button. This will display the aperture selections on your image. Check then uncheck "T1" to make the colors stand out more. Then, go to your image window and click on "File" -> "Save image display as PNG..." You now have a png file of the field image with apertures.

9) You should already have a Dmag vs RMS plot png file and a NEB table txt file (see Section 5.8).

10) Go to "Data Set 2 Fit Settings" window and click on "File" -> "Save image of fit panel as PNG file". This file is not required for the report, but it will be useful when doing a final review of the results with Dr. Plavchan.

11) Type up a brief statement discussing the radius of the photometry aperture in arcseconds, the timing of the transit, transit depths in percentage, etc. See Section D of "TFOP SG1 Guidelines" pdf for further details on that.

12) Do a final review of your results and all aforementioned files above with Dr. Plavchan.

13) If your results are satisfactory, you will need to upload some of the files into the ExoFOP (exofop.ipac.caltech.edu/).

a) If your target's name is "EPIC...", it is generally a K2 target and you will need to upload to ExoFOP-K2.

b) If your target has a TOI ID, it is a TESS target and you will need to upload to ExoFOP-TESS.

c) Some targets are both a K2 and a TESS target. Visit both ExoFOP pages to double-check if they already have a page on a particular target you analyzed and if the target has a TOI ID.

d) Upload the following files to the ExoFOP:

    i) Field image with aperture selections (set it as "Finding_Chart")

    ii) Plate-solved image (set it as "Image")

    iii) Seeing profile

    iv) Measurement table

    v) Plot configuration file

    vi) Light curve plot

    vii) Aperture file

    viii) Dmag plot

    ix) Set files iii through viii as "Photometry".

e) Copy your typed report and paste it into an email, then attach the following files to that email:

    i) Seeing profile

    ii) Plot configuration file

    iii) Light curve plot

    iv) Aperture file

    v) Field image with aperture selections

    vi) Dmag plot

f) Make sure the email's header has the target's name, UT date of observation, the name of the observatory, and the filter used. Send the email.

14) This concludes the preparation and submission of your reports!

8. Additional In-depth PDF Resources for AIJ

AIJ User Guide: www.astro.louisville.edu/software/astroimagej/guide/AstroImageJ_User_Guide.pdf

Transit Light Curve Tutorial: www.astrodennis.com/Guide.pdf

AIJ Cookbook: www.astrodennis.com/AIJCookbook.pdf

TFOP SG1 Guidelines: www.astrodennis.com/TFOP_SG1_Guidelines_Latest.pdf

NEB Table Instructions: www.astrodennis.com/NEB_Table_Instructions.pdf